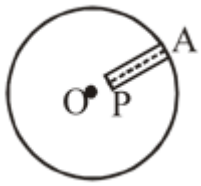
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A planet of radius R has an acceleration due to gravity of g_e on its surface. A deep smooth tunnel is dug on this planet radially inward to reach a point $\mathbf{P}$ located at a distance of $\frac{\mathrm{R}}{2}$ from the centre of the planet. Assume that the planet has uniform density. The kinetic energy required to be given to a small body of mass m, projected radially outward from P, so that it gains a maximum altitude equal to the thrice the radius of the planet from its surface, is equal to



(A) $\frac{63}{16}\mathrm{mgeR}$

(B) $\frac{3}{8}\,\mathrm{mge}\,R$

(C) $\frac{9}{8}\,\mathrm{mge}\,R$

(D) $\frac{21}{16}\mathrm{mgeR}$

Q.2

A body starts from rest from a point distance R_0 from the centre of the earth. The velocity acquired by the body when it reaches the surface of the earth will be (R represents radius of the earth).

(A) $2\mathrm{GM}\left(\frac{1}{\mathrm{R}}-\frac{1}{\mathrm{R}_0}\right)$

(B) $\sqrt{2\mathrm{GM}\left(\frac{1}{\mathrm{R}_0}-\frac{1}{\mathrm{R}}\right)}$

(C) $\mathrm{GM}\left(\frac{1}{\mathrm{R}}-\frac{1}{\mathrm{R}_0}\right)$

(D) $2\mathrm{GM}\sqrt{\left(\frac{1}{\mathrm{R}}-\frac{1}{\mathrm{R}_0}\right)}$

Q.3

A body is projected up with a velocity equal to $3/4$ th of the escape velocity from the surface of the earth. The height it reaches from the centre of the earth is $x\mathrm{R}$. Then what is the value of x ?(Radius of the earth $=\mathrm{R}$)

(A) 2.386

(B) 2.086

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(C) 2.186

(D) 2.286

Q.4

A particle of mass M is situated at the centre of a spherical shell of same mass and radius a. The gravitational potential at a point situated at $\frac{a}{2}$ distance from the centre, will be:

(A) $-\frac{3GM}{a}$

(B) $-\frac{GM}{a}$

(C) $-\frac{4GM}{a}$

(D) $-\frac{2GM}{a}$

Q.5

Assuming the radius of the earth as R, the change in gravitational potential energy of a body of mass m, when it is taken from the earth's surface to a height $3R$ above its surface, is

(A) $3mgR$

(B) $\frac{3}{4}mgR$

(C) $1mgR$

(D) $\frac{3}{2}mgR$

Q.6

The ratio of radii of earth to another planet is $2/\ 3$ and the ratio of their mean densities is $4/5$. If an astronaut can jump to a maximum height of $1.5\ \mathrm{m}$ on the earth, with the same effort, the maximum height he can jump on the planet is

(A) $1\ \mathrm{m}$

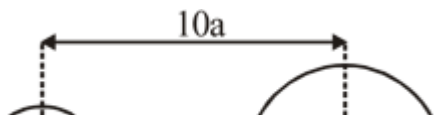
(B) $0.8\ \mathrm{m}$

(C) $0.5\ \mathrm{m}$

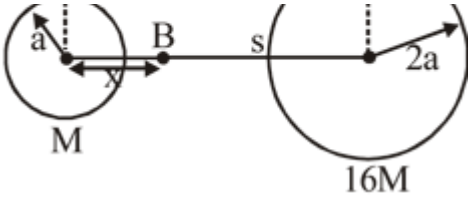
(D) $1.25\ \mathrm{m}$

Q.7

Distance between the centres of two stars is $10\mathrm{a}$. The masses of these stars are M and $16\mathrm{M}$ and their radii a and 2a respectively. A body of mass m is fixed straight from the surface of the larger star towards the smaller star. What should be its minimum initial speed to reach the surface of the smaller star



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(A) $\frac{3}{2}\sqrt{\frac{GM}{a}}$

(B) $\frac{1}{2}\sqrt{\frac{5GM}{a}}$

(C) $\frac{3}{2}\sqrt{\frac{5GM}{a}}$

(D) $\sqrt{\frac{5GM}{a}}$

Q.8

If ' g ' is the acceleration due to gravity on the earth's surface, the gain in the potential energy of an object of mass ' m ' raised from the surface of the earth to a height equal to the radius ' R ' of the earth is

(A) $\frac{1}{4}mgR$

(B) $\frac{1}{2}mgR$

(C) $2\mathrm{mgR}$

(D) mgR

Q.9

A planet in a distant solar system is 10 times more massive than the earth and its radius is 10 times smaller. Given that the escape velocity from the earth is $11\ \mathrm{km\ s^{-1}}$, the escape
velocity from the surface of the planet would be

(A) $11\ \mathrm{km\ s^{-1}}$

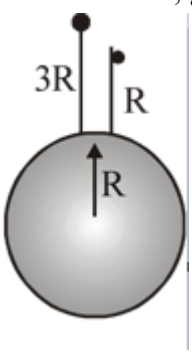
(B) $110\ \mathrm{km\ s^{-1}}$

(C) $0.11\ \mathrm{km\ s^{-t}}$

(D) $1.1\ \mathrm{km\ s^{-1}}$

Q.10

A particle is dropped from height $3\mathrm{R}$ from surface of earth then velocity of particle at height R is (given R is radius of earth, g is acceleration due to gravity) :



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(A) $\frac{\sqrt{gR}}{2}$

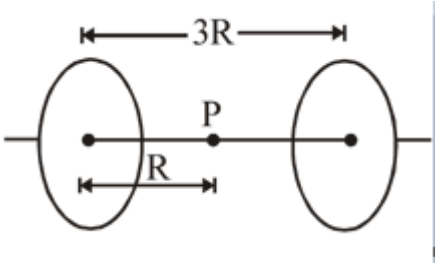
(B) $\frac{2}{3}\sqrt{gR}$

(C) $\sqrt{\frac{gR}{2}}$

(D) none of these

Q.11

Two rings having masses M and $2M$, respectively, having same radius are placed coaxially as shown in figure. If the mass distribution on both the rings is non-uniform, then gravitational potential at point P is



(A) $-\frac{GM}{R}\left[\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}\right]$

(B) $-\frac{\mathrm{GM}}{\mathrm{R}}\left[1+\frac{2}{2}\right]$

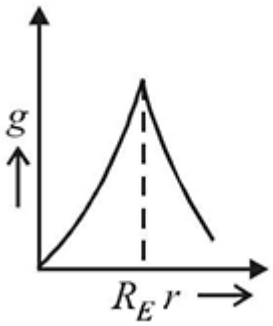
(C) zero

(D) cannot be determined from given information

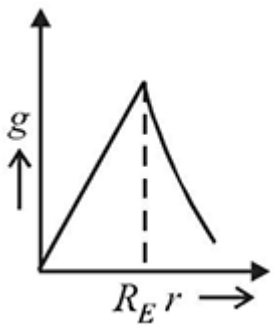
Q.12

Which graph correctly presents the variation of acceleration due to gravity with the distance from the centre of the earth (radius of the earth $=R_E$)?

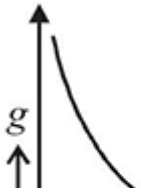
(A)



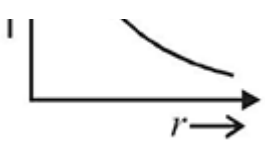
(B)



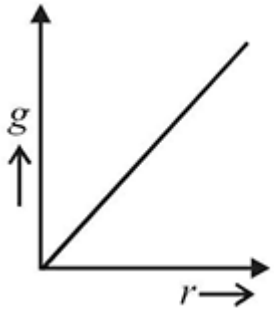
(C)



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(D)



Q.13

Consider Earth to be a homogeneous sphere. Scientist A goes deep down in a mine and scientist B goes high up in a balloon. The gravitational field measured by

(A) A goes on decreasing and that by B goes on increasing

(B) B goes on decreasing and that by A goes on increasing

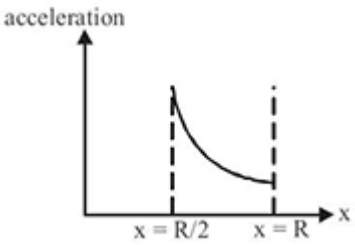
(C) each decreases at the same rate

(D) each decreases at different rates

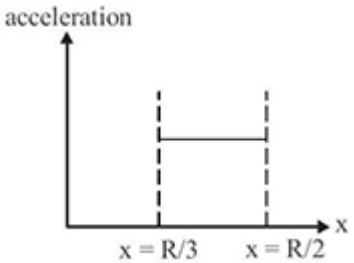
Q.14

A tunnel is dug along a chord of the earth at a perpendicular distance $R/2$ from the earth's centre. The wall of the tunnel may be assumed to be frictionless. A particle is released from one end of the tunnel. The acceleration of the particle varies with x (distance of the particle from the centre) according to acceleration

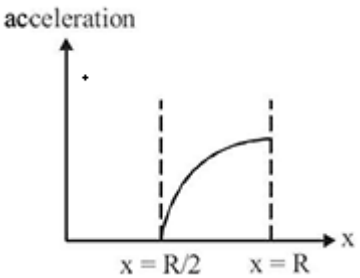
(A)



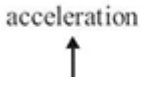
(B)



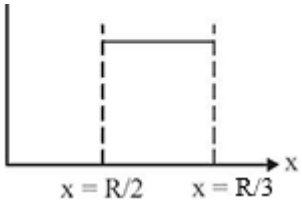
(C)



(D)



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Q.15

The change in the value of ' g ' at a height 'h' above the surface of the earth is the same as at a depth 'd' below the surface of earth. When both 'd' and 'h' are much smaller than the radius of earth, then which one of the following is correct?

(A) $d = \frac{3h}{2}$

(B) $d = \frac{h}{2}$

(C) $d = h$

(D) $d = 2h$

Q.16

Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole $(d << R)$. The value of d is

(A) $\frac{\omega^2 R^2}{g}$

(B) $\frac{\omega^2 R^2}{2g}$

(C) $\frac{2\omega^2 R^2}{g}$

(D) $\frac{\sqrt{Rg}}{g}$

Q.17

Assume that the acceleration due to gravity on the surface of the moon is 0.2 times the acceleration due to gravity on the surface of the earth. If R_e is the maximum range of a projectile on the earth's surface, what is the maximum range on the surface of the moon for the same velocity of projection

(A) $0.2R_e$

(B) $2R_e$

(C) $0.5R_e$

(D) $5R_e$

Q.18

The height of the point vertically above the earth's surface, at which acceleration due to gravity becomes 1% of its value at the earth's surface is (Radius of the earth $= R$)

(A) $8R$

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(B) $9\mathrm{R}$

(C) $10\mathrm{R}$

(D) $20\mathrm{R}$

Q.19

The speed of earth's rotation about its axis is ω. Its speed is increased x times to make the effective acceleration due to gravity equal to zero at the equator. Then x is :

(A) 1

(B) 8.5

(C) 17

(D) 34

Q.20

A uniform ring of mass 2m and radius a is placed directly above the uniform sphere of mass M and of same radius. The centre of ring is at distance $\sqrt{3}$a from the centre of sphere. The gravitational force exerted by the sphere on the ring is N $\frac{\mathrm{GMm}}{\mathrm{a}^2}$ units. Calculate is the value of N ?

(A) 0.41

(B) 0.42

(C) 0.43

(D) 0.44

Answers & Solutions

Q.1 Answer:

$\frac{9}{8}\,\mathrm{mge}\,R$

Solution:

The gravitational potential at a point $Q(OQ=(x)$ is given by :

$$V(x)\left\{\begin{array}{c} =-g_8R\left(\frac{3}{2}-\frac{1}{2}\frac{x^2}{R^2}\right),\ \text{when}\ x<R \\ -g_sR\left(\frac{R}{x}\right),\ \text{when}\ x>R \end{array}\right\}$$

The energy required to project the body, to a maximum altitude of $3\mathrm{R}$ from its surface, is

$$m\left(v_B|_{x-\frac{R}{2}}-V_P|_{x=4R}\right)=\frac{9}{8}m_{i=8}R$$

Q.2 Answer:

$\sqrt{2\mathrm{GM}\left(\frac{1}{\mathrm{R}_0}-\frac{1}{\mathrm{R}}\right)}$

Solution:

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$$\text{P.E.} = \int_{R_0}^{R} \frac{GMm}{r^2} dr = -GMm \left[\frac{1}{R} - \frac{1}{R_0}\right]$$

The K.E. acquired by the body at the surface $= \frac{1}{2} mv^2$

$$\therefore \frac{1}{2}mv^2 = -GMm \left[\frac{1}{R} - \frac{1}{R_0}\right]$$

$$v = \sqrt{2GM\left(\frac{1}{R_0} - \frac{1}{R}\right)}$$

Q.3 Answer:

2.286

Solution:

$v = \frac{3}{4} v_e$

K.E. $= \frac{1}{2}mv^2 = \frac{1}{2}\, m \left(\frac{3}{4}v_e\right)^2 = \frac{9}{32}mv_e^2$

$= \frac{9}{32}\, m \left(\frac{2GM}{R}\right)$

K.E. $= \frac{9}{16}\frac{GMm}{R}$; **P.E.** $= -\frac{GMm}{R}$

Total energy $= \text{K.E.} + \text{P.E.} = -\frac{7}{16}\frac{GMm}{R}$

Let the height above the surface of earth be h**, then P.E.** $= -\frac{GMm}{h}$

$-\frac{7}{16}\frac{GMm}{R} = -\frac{GMm}{h} \quad \therefore h = \frac{16R}{7}$

so $x = \frac{16}{7} = 2.286$

Q.4 Answer:

$-\frac{3GM}{a}$

Solution:

Potential at the given point = Potential at the point due to the shell + Potential due to the particle =

$-\frac{GM}{a} - \frac{2GM}{a} = -\frac{3GM}{a}$

Q.5 Answer:

$\frac{3}{4}mgR$

Solution:

Gravitational potential energy (GPE) on the surface of earth, $E_1 = -\frac{GMm}{R}$

GPE at $3R$, $E_2 = -\frac{GMm}{(R+3R)} = -\frac{GMm}{4R}$

$\therefore$ **Change in GPE** $= E_2 - E_1 = -\frac{GMm}{4R} + \frac{GMm}{R} = \frac{3GMm}{4R}$

$= \frac{3gR^2m}{4R} \left(\because g = \frac{GM}{R^2}\right) = \frac{3}{4}mgR$

Q.6 Answer:

0.8 m

Solution:

K.E. of the person $= \text{P.E.}$ **at the maximum height** $\frac{1}{2}mv^2 = m\frac{GM}{R^2}h$

$\therefore m\frac{GM}{R^2} \cdot h = mG\frac{M'}{R'^2}h'$

Given $\rho = \frac{4}{5}p'$, $\frac{R}{R'} = \frac{2}{3}$

$\therefore \quad mG \cdot \frac{4}{3}\frac{\pi R^3 \rho}{R^2} \cdot h = mG \cdot \frac{4}{3}\frac{\pi R'^3 \rho'}{R'^2} \cdot h'$

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$\Rightarrow \frac{2}{3}R' \cdot \frac{4}{5}\rho' 1.5 = R'\rho' h' \quad \therefore h' = 0.8\ \text{m}$

Q.7 Answer:

$\frac{3}{2}\sqrt{\frac{5GM}{a}}$

Solution:

The distance (from the smaller planet) where the gravitational pulls of the two planets balance each other will be given by

$-\frac{GMm}{x^2} = \frac{-G(16M)m}{(10a-x)^2}$

So, the body will reach the smaller planet due to the planet's gravitational field if it has sufficient energy to cross the point

$B(x = 2a)$ **i.e.** $\frac{1}{2}nv^2 > (v_B - v_s)$

But

$v_a = -\left[\frac{16GM}{2a} + \frac{GM}{(10a-2a)}\right] = -\frac{65GiM}{8a}$

and $v_B = -\left[\frac{16GM}{8a} + \frac{GM}{2a}\right] = -\frac{20GM}{8a}$

So, $\frac{1}{2}mv^2 > m\left[\frac{65GM}{8a} - \frac{20GM}{8a}\right]$

i.e. $V_{min} = \frac{3}{2}\sqrt{\frac{5GM}{a}}$

Q.8 Answer:

$\frac{1}{2}mgR$

Solution:

Gravitational potential energy on the earth surface, $U_r = \frac{-GMm}{R}$

Gravitational potential energy at a height h **above the earth's surface,** $U_h = \frac{-GMm}{R+h}$

$U_h = \frac{-GMm}{R+R} = \frac{-GMm}{2R}$

Gain in gravitational potential energy $= U_h - U_r$

$= \frac{-GMm}{2R} - \left(\frac{-GMm}{R}\right) = \frac{GMm}{R} - \frac{GMm}{2R}$

$= \frac{GMm}{2R} = \frac{1}{2}mgR$

Q.9 Answer:

$110\ \text{km s}^{-1}$

Solution:

$v_p = \sqrt{\frac{2G_0 m_e}{\left(\frac{R}{10}\right)}} = 10\sqrt{\frac{Gm_d}{R}} = 10 \times 11 = 110\ \text{km/sec}$

Q.10 Answer:

$\sqrt{\frac{gR}{2}}$

Solution:

Principle of conservation of energy is to be applied Since acceleration is variable

$\frac{GmM}{4R} = -\frac{GmM}{2R} + \frac{1}{2}mv^2$

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$v = \sqrt{\frac{GM}{2R}} = \sqrt{\frac{gR}{2}}$

Q.11 Answer:

$-\frac{GM}{R}\left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}}\right]$

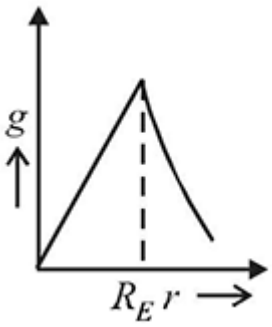
Solution:

As all the points on the periphery of either ring are at same distance from point P, the potential at point P due to whole ring can be calculated as $\mathrm{V} = -(\mathrm{GM})/\left(\sqrt{\mathrm{R}^2 + \mathrm{x}^2}\right)$ where x is axial distance from the centre of the ring. This expression is independent of the fact whether the distribution of mass is uniform or non-uniform.

So, V at P is, $V = \frac{GM}{\sqrt{2R}} - \frac{G \times 2M}{\sqrt{5R}}$

$= -\frac{GM}{R}\left[\frac{1}{\sqrt{2}} + \frac{2}{\sqrt{5}}\right]$

Q.12 Answer:



Solution:

Acceleration due to gravity, $g \propto r$ (if $r < R_E$) and $g \propto \frac{1}{r^2}$ (if $r > R_E$)

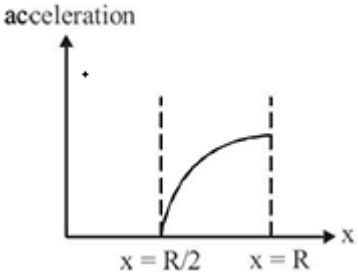
Q.13 Answer:

each decreases at different rates

Solution:

Both decreases but variation are different.

Q.14 Answer:

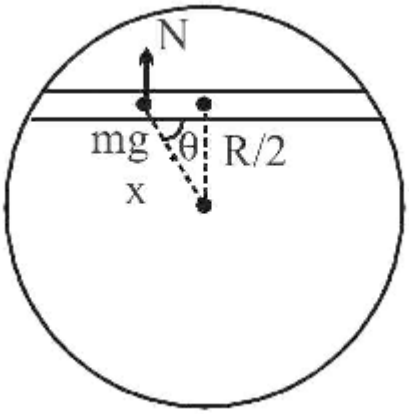


Solution:

Normal force $\mathrm{N} = \mathrm{mg}' \sin\theta$

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Thus pressing force $\mathrm{N}=\frac{\mathrm{mgx}}{\mathrm{R}} \frac{\mathrm{R}}{2 \mathrm{x}}$
$\mathrm{N}=\frac{\mathrm{Mg}}{2}$ **constant and independent of** x **Tangential force,** $\mathrm{F}=\mathrm{ma}=\mathrm{mg}^{\prime} \cos \theta$
$a=\frac{g x}{R} \sqrt{R^2-4 x^2}$
Curve is parabolic and at $\mathrm{x}=\frac{\mathrm{R}}{2}, \mathrm{a}=0$. **Hence** (c).



Q.15 Answer:

$\mathrm{d}=2 \mathrm{~h}$

Solution:

Variation of g with altitude is,
$\mathrm{g_h}=\mathrm{g}\left[1-\frac{2 \mathrm{~h}}{\mathrm{R}}\right]$
variation of g **with depth is,**
$\mathrm{g_d}=\mathrm{g}\left[1-\frac{\mathrm{d}}{\mathrm{R}}\right]$
Equating g_{h} **and** $\mathrm{g_d}$**, we get** $\mathrm{d}=2 \mathrm{~h}$

Q.16 Answer:

$\frac{\omega^2 R^2}{g}$

Solution:

$g\left(1-\frac{d}{R}\right)=g-\omega^2 R; \quad d=\frac{\omega^2 R^2}{g}$

Q.17 Answer:

$5 \mathrm{R_e}$

Solution:

Range of projectile $R=\frac{u^2 \sin 2 \theta}{g}$
if u **and** θ **are constant then** $R \propto \frac{1}{3}$
$\frac{R_m}{R_e}=\frac{g_e}{g_m} \Rightarrow \frac{R_m}{R_e}=\frac{1}{0.2}$
$\Rightarrow R_m=\frac{R_e}{0.2} \Rightarrow R_m=5 R_e$

Q.18 Answer:

$9 \mathrm{R}$

Solution:

$\frac{g^{\prime}}{g}=\left(\frac{R}{R+h}\right)^2 \Rightarrow \frac{1}{100}=\left(\frac{R}{R+h}\right)^2$

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$\Rightarrow h = 9R$

Q.19 Answer:

17

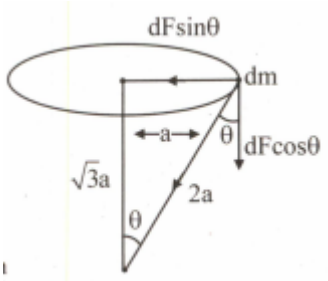
Solution:

$g' = g - \omega'^2 R$, **for** g' **to be zero,**
$\omega' = \sqrt{g/R} \simeq 17\omega$

Q.20 Answer:

0.43

Solution:



$dF = G\frac{Mdm}{4a^2}$

$F = \Sigma\, dF\cos\theta$

$= \sum \frac{GMdm}{4a^2}\cos\theta$

$= \frac{GM}{4a^2} \cdot \frac{\sqrt{3}a}{2a} \sum dm$

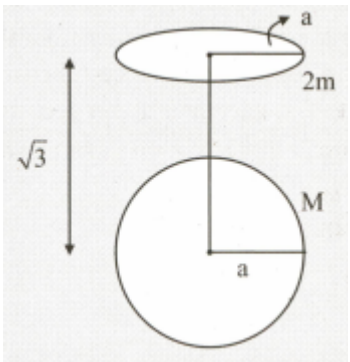
$= \frac{\sqrt{3}GM}{8a^2} \cdot 2m$

$= \frac{\sqrt{3}}{4} \cdot \frac{Gm}{a^2}$

$\therefore \quad F = 0.43\frac{Gm}{a^2}$

$\therefore \quad N = 0.43$

Alternate method



Gravitational field $(\overline{E}) = \frac{G2md}{a^2+d^2{}^{3/2}}$

$= \frac{\sqrt{3}\times 2GM}{8a^2}$

$\therefore \quad \overline{E} = \frac{\sqrt{3}}{4}\frac{Gm}{a^2}$

Force $= M\overline{E} = \frac{\sqrt{3}}{4}\frac{GMm}{a^2}$

$\therefore \quad F = 0.433\frac{GMm}{a^2}$

$N = 0.43$

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